TAXONOMY OF WORK

SKILL EXTRACTION & GOLD-SET EVALUATION

GROUP 3
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TAXONOMY OF WORK CAPSTONE - NORTHEASTERN x WORKHUMAN

(Slide 1 - title)

SLIDE 2

What we're building

Pipeline: raw public job postings -> organised, benchmarked skill taxonomy

- **1: COLLECT** - gather public postings from approved sources (Sprint 1, done)
- **2: EXTRACT - this sprint:** one atomic skill statement per skill, and measure its quality
- **3: ORGANIZE** - next sprints: cluster, label, build the hierarchy, benchmark vs ESCO / O*NET

SLIDE 3

Sprint 2 goals

- **1.** Choose and justify an extraction approach
- **2.** Hand-label 100 postings as a gold set
- **3.** Measure precision / recall, and log LLM token usage

SLIDE 4

One corpus, cleaned - nothing deleted

Kaggle LinkedIn Job Postings (arshkon), Mar-Apr 2024, CC0

	Count
Postings	123,849
Exact duplicate descriptions (tagged)	16,320
Too short (<200 chars) / missing	636 / 7
Usable unique postings	**106,909**

- **Tag, don't drop:** every row stays, with flags (is_duplicate, is_short, is_usable).
- **Line breaks kept:** bullets stay readable for labellers and the LLM; duplicate flags unchanged.
- **Contacts masked:** emails and phone numbers removed before sharing.

SLIDE 5

Gold set: 100 postings, every sector

Sector	Postings
IT	15
Health Care	15
Engineering	12
Finance + Accounting	12
Sales + Marketing	12
Management	10
Legal	8
Admin / HR / Customer Service	8
Mixed	8

- Random, stratified by LinkedIn's first-listed job function (seed 42), 300-5,000 characters, usable rows only.
- LinkedIn sector tags are noisy (an 'Attorney' tagged IT), so we added a corrected sector_fixed.
- Result: **1,317 gold skills**; 1 posting correctly has **zero** skills.

SLIDE 6

What counts as a skill

Clear rules before labelling (data/gold/LABELING_RULES.md)

Label it	Do not label it
Tools: Excel, Salesforce, Epic, Westlaw	Degrees, licences, certifications
Methods: phlebotomy, financial modeling	Years of experience
Knowledge: GAAP, HIPAA compliance	Personality traits ('self-starter')
Soft skills: written communication	Benefits, company info, EEO text
Languages: Spanish	Duties that name no skill ('manage the team')

- **Duty rule (team decision):** a duty that *names* a skill counts - 'build prototypes' -> prototyping.
- One skill per item, posting wording, no filler words ('strong', 'proficiency in').

How we labelled: LLM-suggested, human-decided

- An LLM (Claude / ChatGPT) **suggested** skills; a human **accepted, edited or rejected** each one. Both lists are saved, so human edits are measurable.
- 3 annotators: Priyanka 55 postings - Lochan 35 - Dhanush 10.
- Doubtful skills checked against **ESCO v1.2.1** (13,939 skills) and **O*NET 31.0** (31,821 software/tool rows) with our check_skill.py lookup, plus the web for new tools.
- Finding: ESCO misses many **US legal procedures** (depositions, written discovery) and niche tools - 'not in ESCO' does not mean 'not a skill'.
- **Quality flags reported honestly:** 3 postings double-labelled (exact-name agreement 30-52%, mostly wording); 12 rows kept the AI suggestion unchanged.

Extraction approach: LLM prompting with structured output

```
posting text --> prompt v1 (our labelling rules) --> LLM (JSON: skill + evidence) --> statements
                        cache on disk  |  rate-limit retries  |  token log
```

- **Why an LLM:** skills appear in free text, duties and lists, with many wordings - rules/NER miss most of them.
- **Model:** Qwen 3.8-27B on the **Groq free tier** (brief p.7). Temperature 0, JSON output.
- **Independence:** we avoided OpenAI and Anthropic models, because ChatGPT/Claude suggested the gold labels.
- **Gemini 3.8 Flash** (Google AI Studio) was also run, but returned 'high demand' (503) errors - only 7/100 finished.
- **Engineering:** provider-agnostic backend (one flag), every response cached (re-runs free), retries, keys only in .env.

Results: F1 = 0.72 on 100 hand-labelled postings

Matching	Precision	Recall	F1
Strict (same name)	0.60	0.66	0.63
Lenient (word overlap)	**0.69**	**0.76**	**0.72**

LLM tokens for these scores	Input	Output	Total
All 100 postings	91,554	41,902	133,456
Per posting (average)	916	419	1,335
Per extracted skill (1,448 skills)			~92

- Lenient = one name's words appear in the other ('prototyping' ~ 'high-fidelity prototyping').
- **100%** of evidence phrases are copied exactly from the posting - no invented text.
- Zero-skill posting: the model correctly returned **0** skills.
- Tokens: Qwen 3.8-27B on Groq free tier, $0. Output includes the model's reasoning tokens.

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Results by sector (lenient F1)

Suggested visual: horizontal bar chart

Sector	F1
IT	**0.81**
Engineering	0.75
Admin / HR / Customer Service	0.75
Mixed	0.72
Finance + Accounting	0.71
Legal	0.68
Sales + Marketing	0.67
Management	0.63
Health Care	0.61

- **Best: IT** - tools have clear names (SQL, Snowflake, Jira).
- **Weakest: Health Care, Management** - duties and skills blur ('patient care', 'safety checks').
- Sector counts use sector_fixed, so they differ slightly from slide 5.

Error analysis: where the model and humans disagree

320 missed gold skills - 451 extra predicted skills (lenient)

Error type	Example
Granularity (split vs merged)	gold 'SAP/Chempax/Concur' merged -> model lists SAP, Concur separately
Wording	'preparing discovery' vs gold 'drafting discovery'
Duty vs skill boundary	model adds 'patient care', 'safety checks', 'requirements gathering'
Soft skills / traits	model skips 'organized', 'multitasking'; humans kept them
Gold inconsistency	some gold rows rename or merge skills against our own rules

- Many 'errors' are **gold-set wording**, not model mistakes -> next step: normalise gold names.

LLM usage: free tier, fully logged

	Value
Provider / model	Groq - qwen/qwen3.8-27b
API calls	100 (each posting paid once, then cached)
Input tokens	91,554
Output tokens	41,902
Cost	$0 (free tier)

- Gemini attempt: 7 postings before 503 capacity errors.
- Claude Code was used to build and test the pipeline; usage is documented in the sprint report.

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Limitations & next steps

Limitations

- Gold labels were LLM-suggested; 12 rows unchanged -> possible anchoring.
- Only 3 postings double-labelled -> agreement estimate is weak.
- Kaggle data is 2024; one LinkedIn-derived source.
- Name-based matching (the gold set has no evidence offsets).

Next (Sprint 3)

- Run the extractor on a stratified sample of the corpus -> statements table.
- Normalise gold names; adjudicate double-labelled postings.
- Prompt tuning only on a dev split (to avoid overfitting the 100).
- Embed and cluster statements; map to ESCO / O*NET.

Thank You

(Slide 14)